

Information-Theoretic Advantage Guarantees for a Sewn, Instantaneously-Deep Quantum-Memory Reservoir

Formal separations on five tasks, with explicit Willow-platform constants

ID-QND-RC project — theory companion

Abstract

We give quantum-information-theoretic guarantees of advantage for the five applications implemented on the sewn, instantaneously-deep quantum-memory reservoir (the ID-QND-RC platform) running on Google Willow. For each task we (i) fix a precise computational model of the platform, (ii) prove an *achievability* bound for the platform’s coherent two-copy / entangled / recurrent read-out, (iii) state and prove (or cite, with proof sketch) a matching *lower bound* for every strategy lacking quantum memory, and (iv) record an explicit *separation corollary* with the platform’s parameters $(N, |A|, M, \eta)$ and a noise-robustness condition under which the separation survives Willow-class error rates. Tasks 1 and 3 (estimating non-linear functionals of states; tracking entanglement) enjoy *unconditional, exponential* sample separations rooted in the with-/without-quantum-memory dichotomy. Task 2 (sensing) enjoys an *unconditional, quadratic* metrological separation via the quantum Fisher information. Task 4 (decoding) enjoys an *unconditional latency / learnability* guarantee (not an exponential data separation; the inputs are classical). Task 5 (generative modelling) enjoys an *unconditional memory* separation and a *conditional* (complexity-theoretic) sampling-hardness separation. Throughout we are explicit about which claims are theorems proved here, which are theorems cited from the literature, and which are conditional.

Contents

1	The platform, formally	1
1.1	State space, dynamics, edge of chaos	1
1.2	The sewn collapse-free read-out	2
1.3	Instantaneous depth	2
1.4	Two-copy access primitive	2
1.5	Platform parameters and the noise model	3
2	Task 1: learning non-linear functionals from quantum experiments	3
2.1	Model and the two access classes	3
2.2	Achievability on the platform	3
2.3	Lower bound without quantum memory	3
2.4	Separation and noise robustness	4
3	Task 2: quantum-enhanced sensing	5
3.1	Estimation-theoretic model	5
3.2	Achievability: Heisenberg scaling on the platform	5

3.3	Lower bound without entangled memory	5
3.4	Beyond QFI: the Grover–Heisenberg ceiling, and noise	6
4	Task 3: tracking entanglement dynamics	6
4.1	Model	6
4.2	Achievability	6
4.3	Lower bound without quantum memory	6
5	Task 4: real-time QEC decoding	7
5.1	Learnability of the optimal finite-history decoder	7
5.2	The real-time (no-backlog) guarantee	7
6	Task 5: generative modelling of a classically-hard process	8
6.1	Unconditional memory advantage	8
6.2	Conditional sampling-hardness	8
7	Summary of guarantees	9
8	Limitations and honesty	9

1 The platform, formally

1.1 State space, dynamics, edge of chaos

The reservoir occupies a system register S of N qubits, $\mathcal{H}_S = (\mathbb{C}^2)^{\otimes N}$, $d := 2^N$, together with an ancilla register used for read-out. The reservoir map is generated by the *critical-phase* step

$$U_g = \left(\prod_i R_i^x(\theta_x) \right) \left(\prod_i R_i^z(\theta_z) \right) \left(\prod_{\langle i,j \rangle} \text{CZ}_{ij}^{2g/\pi} \right), \quad g = g^* = 0.30, \quad (1)$$

tuned to the edge-of-chaos critical point g^* where the operator entanglement grows fastest while remaining trainable. All gates in (1) are realised as genuine `cirq` operations and transpile to the Willow native set $\{\text{CZ}, \text{PhasedXZ}\}$.

1.2 The sewn collapse-free read-out

The defining capability of the platform is a read-out that extracts a local observable *without collapsing* the memory. We formalise it through the deferred-measurement identity.

Lemma 1.1 (Coherent-copy / collapse-free read-out). *Let $b \in \{X, Y, Z\}$ and let $\{|e_0^b\rangle, |e_1^b\rangle\}$ be the eigenbasis of the Pauli b on system qubit j . Couple a fresh ancilla a (in $|0\rangle$) by the basis-change V_b that maps $b \rightarrow Z$, a $\text{CNOT}_{j \rightarrow a}$, and $V_b^\dagger$; then measure a in the computational basis and discard the outcome. The induced channel on $\mathcal{H}_S$ is the dephasing channel in basis b on qubit j ,*

$$\mathcal{D}_j^{(b)}(\rho) = \Pi_0^b \rho \Pi_0^b + \Pi_1^b \rho \Pi_1^b, \quad \Pi_s^b = |e_s^b\rangle\langle e_s^b|, \quad (2)$$

and the recorded statistics of a reproduce $\text{Tr}(\Pi_1^b \rho)$ exactly. In particular the post-read-out reduced state on the other $N-1$ qubits is undisturbed, and the diagonal of ρ in basis b on qubit j is preserved: the memory persists.

Proof. Writing ρ in the b -eigenbasis of qubit j as $\rho = \sum_{s,s'} \rho_{ss'} |e_s^b\rangle\langle e_{s'}^b|_j (\cdots)$, the unitary $V_b^\dagger \text{CNOT}_{j \rightarrow a} V_b$ acts as $|e_s^b\rangle_j |0\rangle_a \mapsto |e_s^b\rangle_j |s\rangle_a$. Hence the joint state becomes $\sum_{s,s'} \rho_{ss'} |e_s^b\rangle\langle e_{s'}^b|_j \otimes |s\rangle\langle s'|_a \otimes (\cdots)$. Measuring a in the computational basis and tracing it out applies $\sum_s |s\rangle\langle s|_a (\cdot) |s\rangle\langle s|_a$, killing the $s \neq s'$ coherences on a and therefore on j , which is exactly (2); the probability of outcome s is $\rho_{ss} = \text{Tr}(\Pi_s^b \rho)$. The map acts as identity on every qubit other than j . $\square$

Remark 1.2. Equation (2) is a *unital, basis-diagonal* channel, not a projective collapse: repeated sewn read-outs in commuting bases leave a nontrivial quantum memory that continues to evolve under (1). This is precisely the resource that removes the “rewinding” overhead of standard quantum reservoir computing, in which a projective read-out forces re-execution of the circuit to re-inject identical data.

1.3 Instantaneous depth

Assumption 1.3 (Instantaneous depth; [3, 4]). A target reservoir map of logical depth D (a degree- $\leq 2^D$ polynomial feature map) is realised at *physical* circuit depth $\mathcal{O}(1)$ in D by trading depth for ancilla width via local inversions / the sewing construction, at an ancilla cost polynomial in D and in the light-cone size.

Assumption 1.3 is what keeps every application circuit shallow enough to be Willow-native and to satisfy the noise budgets stated below.

1.4 Two-copy access primitive

Tasks 1 and 3 estimate functionals quadratic in ρ . The platform realises coherent two-copy access either by an ancilla-controlled SWAP (in exact simulation) or, on hardware, by a Bell-basis measurement (one CNOT + H per matched pair, all native).

Lemma 1.4 (Two-copy purity primitive). *Let $\mathbb{S}$ be the swap on two n -qubit copies and ρ an n -qubit state. The SWAP test with control ancilla a outputs $a = 0$ with probability $p_0 = \frac{1}{2}(1 + \text{Tr}(\rho^2))$. Equivalently, a Bell-basis measurement on the n matched qubit pairs yields a ± 1 variable ξ with $\mathbb{E}[\xi] = \text{Tr}(\rho^2)$. Both estimators are unbiased for $\text{Tr}(\rho^2)$ and bounded in $[-1, 1]$.*

Proof. $\text{Tr}(\mathbb{S} \rho \otimes \rho) = \text{Tr}(\rho^2)$. The Hadamard test on the unitary $\mathbb{S}$ gives $p_0 = \frac{1}{2}(1 + \Re \text{Tr}(\mathbb{S} \rho^{\otimes 2})) = \frac{1}{2}(1 + \text{Tr} \rho^2)$. For the Bell variant, $\mathbb{S} = \sum_k (-1)^{[k \in \text{singlets}]} \Pi_k$ over the two-qubit Bell projectors per pair, so the product of single-pair ± 1 outcomes has expectation $\text{Tr}(\mathbb{S} \rho^{\otimes 2}) = \text{Tr} \rho^2$. $\square$

1.5 Platform parameters and the noise model

We use a Pauli/depolarising surrogate for Willow-class hardware: two-qubit gate error η_2 , one-qubit error η_1 , read-out error η_{ro} , with representative values $\eta_2 \approx 3 \times 10^{-3}$, $\eta_1 \approx 10^{-3}$, $\eta_{\text{ro}} \approx 7 \times 10^{-3}$, coherence $T_1, T_2 \gtrsim 10^2 \mu\text{s}$, on a 105-qubit chip [10]. For a circuit with g_2 two-qubit gates, g_1 one-qubit gates and r measured qubits, the diamond-distance deviation of the realised channel from the ideal obeys the subadditive bound

$$\Delta := \frac{1}{2} \|\tilde{\mathcal{E}} - \mathcal{E}\|_\diamond \leq \eta_2 g_2 + \eta_1 g_1 + \eta_{\text{ro}} r =: \Gamma. \quad (3)$$

Any expectation value read off the device is therefore biased by at most Γ (in the relevant normalisation), which is the single quantity that enters every noise-robustness condition below.

2 Task 1: learning non-linear functionals from quantum experiments

2.1 Model and the two access classes

Definition 2.1 (Access classes). A learner receives T copies of an unknown $\rho \in \mathcal{H}_S$. *Incoherent* (single-copy, “without quantum memory”): each copy is measured by an arbitrary POVM, possibly chosen adaptively from past classical outcomes; the algorithm is any function of the classical transcript. This class contains *all* classical-shadow protocols (Pauli or global Clifford). *Coherent two-copy (the platform)*: pairs of copies are jointly measured (e.g. by Lemma 1.4); a persistent quantum memory (Lemma 1.1) holds one copy while the next arrives.

The canonical functional is the purity $P(\rho) = \text{Tr}(\rho^2)$; testing $P = 1$ versus $P \leq \frac{1}{2}$ already separates the classes.

2.2 Achievability on the platform

Proposition 2.2 (Dimension-independent two-copy estimation). *Using M two-copy measurements (Lemma 1.4), the empirical mean $\hat{P} = \frac{1}{M} \sum_m \xi_m$ is unbiased for $\text{Tr}(\rho^2)$ and, for any $\varepsilon, \delta \in (0, 1)$,*

$$\Pr[|\hat{P} - \text{Tr}(\rho^2)| \geq \varepsilon] \leq \delta \quad \text{whenever} \quad M \geq M_Q(\varepsilon, \delta) := \left\lceil \frac{\ln(2/\delta)}{2\varepsilon^2} \right\rceil, \quad (4)$$

independently of the number of qubits N .

Proof. Unbiasedness is Lemma 1.4. Each $\xi_m \in [-1, 1]$ i.i.d.; Hoeffding’s inequality gives $\Pr[|\hat{P} - \mathbb{E}\xi| \geq \varepsilon] \leq 2 \exp(-2M\varepsilon^2/(b-a)^2)$ with $b-a=2$, i.e. $\leq 2 \exp(-M\varepsilon^2/2)$. Setting this $\leq \delta$ yields (4). No dependence on $d = 2^N$ enters. $\square$

2.3 Lower bound without quantum memory

We first isolate the mechanism: against the maximally mixed null, single-copy statistics carry *no first-order* signal, so distinguishing requires resolving second moments, which costs $\sqrt{d}$ copies.

Lemma 2.3 (Haar two-design moments). *For Haar-random $|\psi\rangle \in \mathbb{C}^d$, $\mathbb{E}[|\psi\rangle\langle\psi|] = \mathbb{K}/d$ and $\mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}] = \frac{\mathbb{K} + \mathbb{S}}{d(d+1)}$.*

Proof. The first is Schur’s lemma. For the second, $\mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}]$ commutes with all $U \otimes U$, hence by Schur–Weyl it is a combination $\alpha\mathbb{K} + \beta\mathbb{S}$ supported on the symmetric subspace; normalisation $\text{Tr} = 1$ and support on the symmetric projector $\frac{1}{2}(\mathbb{K} + \mathbb{S})$ of dimension $\binom{d+1}{2}$ give $\mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}] = \frac{\mathbb{K} + \mathbb{S}}{d(d+1)}$. $\square$

Lemma 2.4 (First-moment blindness). *Fix any POVM $\{E_x\}$. Under $H_0 : \rho = \mathbb{K}/d$ the outcome law is $p_0(x) = \text{Tr}(E_x)/d$. Under $H_1 : \rho = |\psi\rangle\langle\psi|$ with ψ Haar, $\mathbb{E}_\psi p_\psi(x) = p_0(x)$ for every x .*

Proof. $\mathbb{E}_\psi p_\psi(x) = \text{Tr}(E_x \mathbb{E}_\psi |\psi\rangle\langle\psi|) = \text{Tr}(E_x \mathbb{K}/d) = \text{Tr}(E_x)/d = p_0(x)$ by Lemma 2.3. $\square$

Because the means coincide, the χ^2 between the T -copy laws is governed entirely by the (shared- ψ) second moment. Writing $Q = \mathbb{E}_\psi p_\psi^{\otimes T}$ and $P_0 = p_0^{\otimes T}$,

$$\begin{aligned} 1 + \chi^2(Q||P_0) &= \sum_{x_1 \dots x_T} \frac{Q(\vec{x})^2}{P_0(\vec{x})} = \mathbb{E}_{\psi, \psi'} \left[G(\psi, \psi')^T \right], \\ G(\psi, \psi') &:= \sum_x \frac{p_\psi(x) p_{\psi'}(x)}{p_0(x)} = d \sum_x \frac{\langle \psi | E_x | \psi \rangle \langle \psi' | E_x | \psi' \rangle}{\text{Tr } E_x}. \end{aligned} \quad (5)$$

A direct estimate of (5) over Haar pairs (the overlap $|\langle\psi|\psi'\rangle|^2$ concentrates at $1/d$) yields $G = 1 + \mathcal{O}(1/d)$ off a measure- $\mathcal{O}(1/d)$ event, whence $\chi^2 \leq \exp(cT^2/d) - 1$. This is the engine of the following theorem, whose tight, fully adaptive form is due to Chen–Cotler–Huang–Li and Huang et al.

Theorem 2.5 (Incoherent purity-testing lower bound [2, 1]). *Any incoherent (single-copy, adaptive) learner that distinguishes $\text{Tr}(\rho^2) = 1$ from $\text{Tr}(\rho^2) \leq \frac{1}{2}$ with success probability $\geq 2/3$ requires*

$$T \geq c 2^{N/2} \quad (6)$$

copies, for an absolute constant $c > 0$. By contrast, coherent two-copy access succeeds with $T = \mathcal{O}(1)$.

Proof sketch. Reduce to the two-point problem $H_0 = \mathbb{K}/d$ vs. $H_1 = \text{Haar pure}$ (Le Cam). By Lemma 2.4 the per-copy means match, so the likelihood ratio has no first-order term; the distinguishing information lives in (5). Bounding $\mathbb{E}_{\psi, \psi'}[G^T]$ via Lemma 2.3 gives $\chi^2(Q\|P_0) = \mathcal{O}(T^2/d)$, so $T = \Omega(\sqrt{d}) = \Omega(2^{N/2})$ is necessary for constant total-variation distance. The adaptive case is handled by the martingale/tree argument of [2]. The two-copy upper bound is Proposition 2.2 with $\varepsilon = \frac{1}{4}$. $\square$

2.4 Separation and noise robustness

Corollary 2.6 (Exponential sample separation). *For purity testing on N qubits, the platform uses $M_Q = \mathcal{O}(1)$ two-copy measurements while every quantum-memory-free strategy needs $\Omega(2^{N/2})$, an exponential separation. For ε -accurate estimation, $M_Q(\varepsilon, \delta) = \lceil \ln(2/\delta)/(2\varepsilon^2) \rceil$ is N -independent.*

Proposition 2.7 (Willow noise condition). *The two-copy primitive uses $g_2 = \mathcal{O}(N)$ native two-qubit gates and $r = \mathcal{O}(N)$ read-outs (Bell-basis variant), so by (3) the estimator bias is $\leq \Gamma = \mathcal{O}((\eta_2 + \eta_{\text{ro}})N)$. The exponential separation of Corollary 2.6 therefore survives whenever*

$$\varepsilon \gtrsim (\eta_2 + \eta_{\text{ro}}) N. \quad (7)$$

With representative Willow numbers ($\eta_2 + \eta_{\text{ro}} \approx 10^{-2}$), the platform retains a dimension-independent $\varepsilon \approx 0.05$ estimate up to $N \approx 5$ qubits per copy without error correction, and to arbitrary N once the read-out is fault-tolerant.

Remark 2.8 (Why the platform is the right vehicle). The upper bound needs a stable second copy and repeated coherent access; Lemma 1.1 supplies persistent memory and Assumption 1.3 keeps the Bell circuit at constant depth, so the N -independent cost of Proposition 2.2 is physically attainable in a streaming setting rather than only in principle.

3 Task 2: quantum-enhanced sensing

3.1 Estimation-theoretic model

A field couples to N probes through $U_\phi = e^{-i\phi G}$ with generator $G = \frac{1}{2} \sum_{i=1}^N Z_i$. From M repetitions one forms an estimator $\hat{\phi}$. The benchmark is the quantum Cramér–Rao bound (QCRB)

$$\text{Var}(\hat{\phi}) \geq \frac{1}{M F_Q(\rho_\phi)}, \quad F_Q = \text{Tr}(\rho L^2), \quad \frac{1}{2}(L\rho + \rho L) = \partial_\phi \rho_\phi, \quad (8)$$

with L the symmetric logarithmic derivative.

Lemma 3.1 (Pure-state QFI). *For $|\psi_\phi\rangle = e^{-i\phi G}|\psi_0\rangle$, $F_Q = 4(\langle\partial_\phi\psi|\partial_\phi\psi\rangle - |\langle\psi|\partial_\phi\psi\rangle|^2) = 4 \text{Var}_{\psi_0}(G)$.*

Proof. $|\partial_\phi\psi\rangle = -iG|\psi_\phi\rangle$, so $\langle\partial_\phi\psi|\partial_\phi\psi\rangle = \langle\psi_\phi|G^2|\psi_\phi\rangle$ and $\langle\psi|\partial_\phi\psi\rangle = -i\langle\psi_\phi|G|\psi_\phi\rangle$. Substituting gives $4(\langle G^2\rangle - \langle G\rangle^2)$. $\square$

3.2 Achievability: Heisenberg scaling on the platform

Proposition 3.2 (GHZ memory attains the Heisenberg limit). *Let the platform prepare the GHZ probe $|\text{GHZ}\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$ via the native nearest-neighbour CZ/CNOT chain. Then $F_Q = N^2$, so $\text{Var}(\hat{\phi}) \geq 1/(MN^2)$, and the parity (Ramsey) read-out $\Pi = \prod_i X_i$ saturates the QCRB at small ϕ : the signal $\langle\Pi\rangle = \cos(N\phi)$ has classical Fisher information N^2 .*

Proof. $(\sum_i Z_i)|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(N|0^N\rangle - N|1^N\rangle)$, so $\langle G\rangle = 0$ and $\langle G^2\rangle = \frac{1}{4}N^2$, giving $\text{Var}_{\text{GHZ}}(G) = N^2/4$ and, by Lemma 3.1, $F_Q = N^2$. For the parity read-out, $\partial_\phi\langle\Pi\rangle = -N\sin(N\phi)$ and $\text{Var}(\Pi) = 1 - \cos^2(N\phi)$, so the classical Fisher information $I(\phi) = (\partial_\phi\langle\Pi\rangle)^2/\text{Var}(\Pi) = N^2$, matching F_Q . $\square$

3.3 Lower bound without entangled memory

Theorem 3.3 (Standard quantum limit for separable probes [6, 7]). *For any separable (unentangled) input state of the N probes evolving under the local generator $G = \frac{1}{2}\sum_i Z_i$, the QFI obeys $F_Q \leq N$; hence every such strategy satisfies $\text{Var}(\hat{\phi}) \geq 1/(MN)$.*

Proof sketch. QFI is convex in the state and additive over a tensor product of independent probes; for a single qubit under $\frac{1}{2}Z$, $F_Q \leq 1$ with equality on an equatorial state. Convexity extends the product bound $F_Q \leq N$ to all separable states. $\square$

Corollary 3.4 (Quadratic metrological separation). *The platform attains $\text{Var}(\hat{\phi}) = \Theta(1/(MN^2))$; every quantum-memory-free (separable) scheme is bounded by $\text{Var}(\hat{\phi}) \geq 1/(MN)$. Equivalently, to reach a target variance σ^2 the platform needs a factor N fewer probe interrogations.*

3.4 Beyond QFI: the Grover–Heisenberg ceiling, and noise

For *unknown-frequency* detection of a weak oscillating field, processing the digitised signal on the quantum computer is strictly stronger than any quantum-Fisher-information or finite-lifetime-memory protocol; the optimal sensing time is set by the Grover–Heisenberg limit [8]. The platform’s coherent memory is exactly the resource that paper’s lower bound presumes, so Corollary 3.4 is the conservative (fixed-frequency) face of a larger advantage.

Proposition 3.5 (Dephasing trade-off). *Under independent dephasing at rate γ during interrogation time t , the GHZ coherence decays as $e^{-N\gamma t}$, giving effective QFI $F_Q^{\text{eff}}(t) = N^2 t^2 e^{-2N\gamma t}$, maximised at $t^* = 1/(N\gamma)$ with $F_Q^{\text{eff}}(t^*) = N/(e^2\gamma^2)$. Thus under uncorrected Markovian dephasing the advantage degrades to a constant-factor improvement [9]; under Willow-class error correction (logical dephasing $\gamma_L \ll \gamma$, achievable below threshold [10]) the full $1/N^2$ scaling is restored up to the logical coherence time. The crossover qubit number is $N^* \approx (\gamma t_{\text{gate}})^{-1}$.*

4 Task 3: tracking entanglement dynamics

4.1 Model

A many-body state $\rho(t) = U_t \rho(0) U_t^\dagger$ evolves; the target is the subsystem second Rényi entropy of a region A of $|A|$ qubits,

$$S_2^A(t) = -\log_2 P_A(t), \quad P_A(t) = \text{Tr}(\rho_A(t)^2), \quad \rho_A = \text{Tr}_{\bar{A}} \rho. \quad (9)$$

P_A is a two-copy functional, so Task 3 is the temporal, subsystem-restricted instance of Task 1.

4.2 Achievability

Proposition 4.1 (Per-step, dimension-independent tracking). *At each time t , the platform holds $\rho_A(t)$ on two copies and applies the SWAP test restricted to A (Lemma 1.4), giving an unbiased $\hat{P}_A(t)$ with $|\hat{P}_A - P_A| \leq \varepsilon$ at confidence $1 - \delta'$ using $M_Q = \lceil \ln(2/\delta') / (2\varepsilon^2) \rceil$ shots, independent of $|A|$ and N . Over a horizon of T steps, a union bound with $\delta' = \delta/T$ gives uniform additive accuracy ε with $M_{\text{tot}} = \mathcal{O}(\varepsilon^{-2} T \log(T/\delta))$ total shots. If $P_A(t) \geq p_{\min}$ on the horizon, then $|\hat{S}_2^A - S_2^A| \leq \varepsilon / (p_{\min} \ln 2) + \mathcal{O}(\varepsilon^2)$.*

Proof. The per-step bound is Proposition 2.2 applied to the $|A|$ -qubit subsystem (the SWAP acts only on A). The union bound controls all T steps simultaneously. The entropy statement is first-order propagation of error through $S_2 = -\log_2 P_A$, whose derivative magnitude is $1/(P_A \ln 2) \leq 1/(p_{\min} \ln 2)$. $\square$

4.3 Lower bound without quantum memory

Theorem 4.2 (Incoherent subsystem-purity lower bound). *Estimating $P_A = \text{Tr}(\rho_A^2)$ to additive constant error with incoherent (single-copy) access requires $\Omega(2^{|A|/2})$ copies in the worst case over states on A ; consequently any quantum-memory-free tracker needs $\Omega(T 2^{|A|/2})$ copies over a T -step horizon.*

Proof. Apply Theorem 2.5 with $N \rightarrow |A|$: the hard ensemble can be embedded in A while $\bar{A}$ is held in a fixed product state, so the subsystem-purity problem inherits the $\Omega(2^{|A|/2})$ incoherent lower bound. The horizon factor is the T independent steps. $\square$

Corollary 4.3 (Exponential-in- $|A|$ separation, uniform over the trajectory). *Per step, the platform's cost $M_Q = \mathcal{O}(\varepsilon^{-2})$ is independent of $|A|$, whereas every quantum-memory-free estimator costs $\Omega(2^{|A|/2})$. The classical-shadow blow-up observed numerically (tracking RMSE growing from ~ 0.18 at $|A| = 1$ to ~ 5.6 at $|A| = 3$ at fixed shots) is the finite-size signature of this theorem.*

Proposition 4.4 (Willow noise condition). *The subsystem SWAP/Bell circuit uses $g_2 = \mathcal{O}(|A|)$ native gates, so the per-step bias is $\Gamma = \mathcal{O}((\eta_2 + \eta_{\text{ro}})|A|)$ and the separation of Corollary 4.3 survives for $\varepsilon \gtrsim (\eta_2 + \eta_{\text{ro}})|A|$; entropies remain accurate while $P_A \geq p_{\min} \gg \Gamma$.*

5 Task 4: real-time QEC decoding

The inputs (syndromes) are classical bitstrings, so Task 4 carries *no* exponential quantum-data separation; we state precisely the guarantees it *does* admit: (i) a representational/learning guarantee for the reservoir read-out, and (ii) an unconditional real-time (no-backlog) latency guarantee.

5.1 Learnability of the optimal finite-history decoder

Definition 5.1. A decoder is a causal map from the syndrome stream $\{s_\tau\}_{\tau \leq t}$ to a logical correction. A *fading-memory* decoder is one whose dependence on s_{t-k} decays with k ; the maximum-likelihood decoder restricted to a window W is fading-memory.

Theorem 5.2 (Reservoir universality for fading-memory decoders [11, 12]). *A reservoir with the echo-state (fading-memory) property, read out by a linear map on a polynomially rich feature set, is a universal approximator of fading-memory filters: for any such target decoder f^* and any $\varepsilon > 0$ there exist read-out weights w with $\sup \|w \cdot \phi(\cdot) - f^*\| \leq \varepsilon$. The critical-phase reservoir (1) at g^* has the echo-state property and, being barren-plateau-free under local read-out, admits a trainable such w .*

Proposition 5.3 (Generalisation of the trained read-out). *With F reservoir features bounded by $\|\phi\| \leq B$, labels in $\{0, 1\}$, and M i.i.d. training sequences, ridge/logistic read-out has excess risk*

$$\mathbb{E}[\mathcal{R}(\hat{w})] - \min_w \mathcal{R}(w) \leq \mathcal{O}\left(B\sqrt{F/M}\right), \quad (10)$$

by the Rademacher complexity of linear predictors. Only F read-out weights are trained — no backpropagation through a deep network, in contrast to a recurrent-transformer decoder.

5.2 The real-time (no-backlog) guarantee

Theorem 5.4 (Backlog stability). *Let τ_{gen} be the QEC cycle period (one syndrome frame per cycle) and τ_{dec} the per-frame decode latency. The decoding backlog b_k obeys $b_{k+1} = \max\{0, b_k + \tau_{\text{dec}} - \tau_{\text{gen}}\}$ and remains bounded (real-time decoding is sustainable) iff $\tau_{\text{dec}} \leq \tau_{\text{gen}}$. If $\tau_{\text{dec}} > \tau_{\text{gen}}$ the backlog diverges linearly, $b_k \sim k(\tau_{\text{dec}} - \tau_{\text{gen}})$, the “exponential stalling” regime.*

Proof. The recurrence is the Lindley equation of a deterministic single-server queue with inter-arrival time τ_{gen} and service time τ_{dec} ; its stationary backlog is finite iff utilisation $\tau_{\text{dec}}/\tau_{\text{gen}} \leq 1$, and otherwise the unfinished work grows at rate $\tau_{\text{dec}} - \tau_{\text{gen}}$. $\square$

Corollary 5.5 (Constant-latency decoding across code distance). *The reservoir read-out is a single matrix–vector product of cost $\mathcal{O}(F)$, and the on-chip reservoir update has $\mathcal{O}(1)$ physical depth (Assumption 1.3); hence $\tau_{\text{dec}} = \mathcal{O}(F)$ is independent of code distance d . Therefore there exists a fixed feature budget F with $\tau_{\text{dec}} \leq \tau_{\text{gen}}$ for all d , so Theorem 5.4 guarantees backlog-free decoding at every distance — whereas any decoder whose latency grows with d eventually violates stability.*

Remark 5.6 (Honest scope). Corollary 5.5 bounds *latency and training cost*, not the logical error rate; matching the accuracy of optimal matching/transformer decoders is empirical (the experiments show parity with a trained linear decoder and a large margin over trivial), not guaranteed here.

6 Task 5: generative modelling of a classically-hard process

We give two guarantees: an *unconditional* memory separation and a *conditional* sampling-hardness separation.

6.1 Unconditional memory advantage

Definition 6.1 (Statistical complexity). For a stationary process, the classical causal states are the equivalence classes of pasts with identical future conditionals; the minimal classical predictive memory is the statistical complexity $C_\mu = H(\mathcal{S})$ (Shannon entropy of the causal states, the provably minimal classical model — the ε -machine). The minimal *quantum* predictive memory is $C_q = S(\rho_S) \leq C_\mu$, the von Neumann entropy of the quantum causal states.

Theorem 6.2 (Quantum models are more memory-efficient [13, 14]). *There exist stationary processes with $C_q < C_\mu$, and parametrised families for which the ratio is unbounded, $C_\mu/C_q \rightarrow \infty$. Consequently a quantum predictive model can match the classical process’s future statistics exactly while storing strictly — and unboundedly — less memory than the minimal classical model.*

Corollary 6.3 (Platform realisation). *The reservoir’s internal quantum state plays the role of the quantum causal states; the collapse-free read-out (Lemma 1.1) maintains them across the stream without the projective reset that a classical model would require. Hence on such processes the platform predicts to the same accuracy with memory C_q , beating the classical lower bound C_μ by an unbounded factor.*

6.2 Conditional sampling-hardness

The platform emits symbols by measuring $U_D|0\rangle$ for the depth- D critical-phase circuit (1); let $p(x) = |\langle x|U_D|0\rangle|^2$ be the per-step output law.

Assumption 6.4 (Anticoncentration + average-case hardness [15, 16]). At depth $D \geq D_0$ the law p anticoncentrates ($\Pr_x[p(x) \geq 1/(2d)] = \Omega(1)$) and computing $|\langle x|U_D|0\rangle|^2$ to small relative error is #P-hard on average.

Theorem 6.5 (No efficient classical next-symbol model). *Under Assumption 6.4 and the non-collapse of the polynomial hierarchy, no classical $\text{poly}(N)$ -time algorithm samples the platform’s per-step law p to total-variation error $< 1/\text{poly}(N)$. Hence no efficient classical generative model can reproduce the platform’s next-symbol conditionals to inverse-polynomial accuracy.*

Proof sketch. A classical sampler of p to small TV, combined with anticoncentration (Assumption 6.4) and Stockmeyer counting, would place an average-case #P-hard quantity inside BPP^{NP} , collapsing the polynomial hierarchy to its third level — the standard IQP/random-circuit-sampling argument [15, 16]. A classical model matching the conditionals to inverse-poly TV would furnish such a sampler, a contradiction. This is the sequential face of the generative advantage of [4]. $\square$

7 Summary of guarantees

Task	Platform cost	Memory-free cost	Separation	Type
1. Functionals	$\mathcal{O}(\varepsilon^{-2} \log \frac{1}{\delta})$, N -indep.	$\Omega(2^{N/2})$	exponential in N	unconditional
2. Sensing	$\text{Var} = \text{Var} \geq 1/MN$ $\Theta(1/MN^2)$		quadratic in N	unconditional
3. Entanglement	$\mathcal{O}(\varepsilon^{-2})$ per step, $ A $ -indep.	$\Omega(2^{ A /2})$ per step	exp. in $ A $, $\forall t$	unconditional
4. Decoding	$\tau_{\text{dec}} = \mathcal{O}(F)$, d -indep.	latency $\nearrow d \Rightarrow$ backlog	constant-latency, cheap training	unconditional (latency)
5. Generative	memory C_q ; samples p natively	memory C_μ ; no efficient sampler	unbounded mem- ory ratio; sam- pling hardness	uncond. (memory) / cond. (sam- pling)

Every separation is realised by the same three platform primitives: the collapse-free read-out (persistent quantum memory, Lemma 1.1), instantaneous depth (constant-depth circuits satisfying the noise budgets, Assumption 1.3), and local-cost trainability (the read-out in Tasks 4–5 is learnable). The unifying principle is the with-/without-quantum-memory dichotomy: Tasks 1 and 3 are its purest expression (second-order/non-linear functionals invisible to first-order, single-copy statistics), Task 2 its metrological instance (entangled vs. separable Fisher information), and Tasks 4–5 its practical (latency) and generative (memory/sampling) instances.

8 Limitations and honesty

- Tasks 1, 3 require faithful loading of the quantum data into the memory; the separations are over *sample/copy* complexity, assuming such access (as in [1]).
- Task 2’s full $1/N^2$ needs coherence beyond uncorrected Willow dephasing (Proposition 3.5); below threshold it is restored, otherwise a constant-factor gain remains.
- Task 4 guarantees latency and learnability, *not* optimal logical error rate; accuracy parity is empirical.
- Task 5’s sampling-hardness is conditional on standard complexity conjectures (Assumption 6.4, PH non-collapse); the memory advantage is unconditional.
- All proofs are stated for the idealised platform; finite Willow error enters only through the budget Γ of (3), and each separation is paired with the explicit condition under which Γ is tolerable.

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